Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/07/2008
PI Name:	Feke, Donald L. Ph.D.		
Project Title:	Coalescence, Transport, and Separation of Gas Bubbles in Liquids Using Ultrasonic Processing Techniques		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Multiphase flow and heat transfer		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106 - 7217	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03-C
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	One of the co-investigators on this project (John Graf) left his position at NASA shortly after this award was received. Therefore, the portions of the project originally intended for that subcontract were eliminated from the scope of work of this project.		
COI Name (Institution):	Nakagawa, Masami (Colorado School of Mines)		
Grant/Contract No.:	NNC05GA29G		
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Task Description:

Processes involving liquid flow or storage have the potential to entrain gases or to form bubbles from dissolved gases contained within the liquid. In ground-based research, the significant density difference between dispersed bubbles and the continuous liquid phase enables relatively easy phase separation to occur using conventional means such as vents or traps. However, for microgravity environments, more sophisticated methods must be employed. Rotating equipment or devices that produce swirling flows often require complicated equipment which is often energy inefficient, or flow configurations that are difficult to maintain in stable operation.
In our laboratory, we have developed robust, energy efficient techniques that utilize resonant ultrasonic fields to accomplish control and transport of suspended solid particulates or emulsified droplet phases within their host liquids. Such acoustic methods are easily controllable, energy efficient, and since they are electrical in nature and have no moving mechanical parts, these techniques are quite robust.

Here, we propose to extend these techniques for the phase separation and control of bubble populations within liquids. We will develop and benchmark a method for bubble removal from liquids that uses resonant acoustic fields to induce the coalescence of the bubbles and to drive the transport of bubbles to locations where they can be readily harvested from the liquid. In support of the development of this method, we plan to obtain some fundamental information on the physics that underlies the acoustically driven coalescence of bubbles (both in the bulk liquid and on solid surfaces) and the transport of bubbles within liquids (or the retention of gas bubbles within flowing liquids). Models developed from these analyses will be used to benchmark the performance achieved in ground-based prototype devices. These models will also be useful for the design of potential flight apparatus.

The specific aims of this project will be:

· to design, implement, and evaluate prototype systems to collect and remove bubbles from a gas-liquid dispersion using acoustic fields · to obtain fundamental information on the role of primary and secondary acoustic forces on the coalescence of gas bubble within bulk fluids and adhering to solid surfaces · to perform a modeling study that will focus on the transport and interaction behavior of individual and pairs of gas bubbles suspended in liquids subjected to conditions of mild acoustic resonance · to conduct experiments to validate the outcome of the predictions of acoustic forces on individual bubbles and pairs of bubbles · to design, construct and evaluate the performance of prototype devices in ground-based experiments. · to verify the operation of such devices in microgravity environments such as those achieved in drop-tower experiments or in KC-135 flights

This research will be a collaboration between Prof. Donald L. Feke (Case Western Reserve Univeristy), Dr. Masami Nakagaws (Colorado School of Mines, and NASA Glenn Research Center), and Dr. John Graf (NASA Johnson Space Center). Prof. Feke has expertise in the development and use of acoustic techniques for chemical and materials processing applications. Dr. Nakagawa has considerable experience in the design and conduction of microgravity experiments in both the NASA drop-tower and KC-135 aircraft. Dr. Graf has had more than a decade of experience as a research engineer in the Life Support Systems branch at NASA. The complementary interests and expertise of the three co-investigators will lead to a fruitful collaboration on this project.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This research aimed to develop a fundamental understanding of the effect of forces induced by resonant ultrasonic fields on the on the entrapment and coalescence phenomena of gas bubbles, both in normal and microgravity conditions. Modeling efforts illustrate the relationship between the size of bubbles, their sphericity, acoustic field parameters (frequency and intensity), bubble placement relative to acoustic pressure antinodes, and the bubble coalescence phenomena. Experiments involving small numbers of bubbles (less than ten) are used to test the basic predictions of the model. These results support subsequent studies of the clustering and coalescence of swarms of bubbles.
Besides having direct impact on the design of apparatus to accomplish the separation of gas bubbles from liquid streams in microgravity environments, there are many earth-based applications for the results of this study. For example, the separation of neutrally buoyant (or near neutrally buoyant) solids or emulsified droplets from their host liquids is encountered in a variety of significant and strategic applications. These range from biomedical and biotechnical applications (e.g., the sorting of cells for medical diagnostics, the enhancement of the overall viability of cells within bioreactors) to advanced materials applications (e.g., the creation of composite materials with tailored microstructures) to environmental applications (e.g., the remediation of wastewater streams). All of these applications (and others) could benefit from an advanced understanding of acoustic processing techniques for dispersed phase manipulation.

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This research comprises four tasks:

Task 1: Analysis of the conditions under which ultrasonic fields can be used to entrap individual gas bubbles under normal gravity conditions. Fundamental information obtained from this analysis is used to determine the experimental conditions under which ground-based experiments can be used to study bubble coalescence phenomena induced by acoustics. As part of this task, we also describe the range of acoustic field intensity that can be used without distorting bubbles from sphericity.

Task 2: Experimental verification of the acoustic entrapment phenomena for individual bubbles under normal gravity conditions. Included under this task is an evaluation of the quality of the acoustic field produced in our experimental apparatus.

Task 3: Analysis of the interactions between pairs of bubbles subjected to a resonant ultrasonic field. This modeling study accounts for interaction forces (acoustic, van der Waals, and hydrodynamic) that drives bubbles toward clustering or coalescence following entrapment. Predictions of the kinetics of the clustering phenomena are developed as a

Task Progress:	function of various system parameters (acoustic field intensity, bubble size, etc.). Task 4: Experimental verification of the modeling predictions of bubble interactions under normal gravity conditions. Included here are studies of the different configurations of bubble clusters observed to result from acoustic forces. Significant progress has been made in each task. It was shown that bubbles of micron size could be captured and retained in normal gravity utilizing easily accessible acoustic conditions. Bubble entrapment at discrete positions (near acoustic pressure antinodes) was both modeled and experimentally verified. At these positions, the body forces (e.g., the primary acoustic force and gravity/buoyancy forces) acting on bubbles are in balance. The model allows prediction of conditions for entrapment and kinetics of the transport of individual bubbles to these entrapment positions as a function of acoustic frequency and intensity, bubble size, and fluid properties. The predictive model readily extends to microgravity conditions. Experiments on single bubbles were conducted under conditions for which the acoustic field did not significantly distort bubbles from a spherical shape. Entrapment of bubbles near acoustic pressure antinodes has been verified using imaging data. Extension of the model to understand bubble interaction forces (e.g., secondary acoustic forces, 28 van der Waals interactions) was also accomplished. This analysis enables the prediction of relative bubble motion, which is the first step toward understanding coalescence and clustering kinetics. The model is sensitive to acoustic field and bubble properties. An interesting result from the modeling was the prediction that the time for bubble collision exhibited a maximum with respect to the ratio of the sizes of the two bubbles. Experiments to investigate the transport time of pairs of bubbles (or pairs of clusters) to collision were performed. Measured interaction times were consistent with the results obtained from the modeling studies. This provides confidence that the models developed in this work can be utilized for the design of acoustically driven bubble-separation devices for microgravity environments. The original work statement anticipated that ground-based microgravity experiments would be conducted as part of this work. Unfortunately, due to the truncation of this grant, those objectives were removed from the work plan. However, the work reported here forms the foundation for taking those next steps in a future research project. Both the modeling and experimental studies can readily be extended to anticipate the transport and coalescence behavior of bubbles subjected to an acoustic field under microgravity conditions. In addition, two papers are being submitted to Chemical Engineering Science and/or Ultrasonics.
Bibliography Type:	Description: (Last Updated: 02/08/2008)
Papers from Meeting Proceedings	Malers J, Feke DE. "Fundamentals of Bubble Transport in an Acoustically Assisted Separation Process." AIChE Annual Meeting, San Francisco, November, 2006. AIChE Annual Meeting, San Francisco, November, 2006. Conference proceedings on CD-ROM, paper #234f. ISBN 0-8169-1012-X , Nov-2006